

Gavindya Jayawardena

School of Information
The University of Texas at Austin
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RESEARCH INTERESTS

Eye-tracking, Interactive-Information Retrieval, Human-Computer Interaction, Data Science, Machine Learning

EDUCATION

Sept. 2024 -
Present

Postdoctoral Scholar, School of Information, The University of Texas at Austin, Austin, TX
Bullard Postdoctoral Research Fellowship
Early Career Scholars Postdoctoral Fellow (2025-2026)
Supervisor/Mentor: *Associate Professor Jacek Gwizdka*
Mentor: *Professor James Howison*

Jan. 2019 -
Aug. 2024

Ph.D., Computer Science, Old Dominion University, Norfolk, VA
Dissertation: *"RAEMAP: Real-Time Advanced Eye Movements Analysis Pipeline"*
Advisor: *Associate Professor Sampath Jayarathna*
Committee Members: *Professor Michele C. Weigle, Professor Michael L. Nelson, Assistant Professor Vikas G. Ashok, Associate Professor Anne M. Perrotti*
Courses: *Data Science and Analytics, Machine Learning, Practical Machine Learning and Applications, Information Retrieval, Algorithms and Data Structures, DevOps & Containers, Natural Language Processing, Blockchains and Cryptocurrencies, Deep Learning on Graphs*

Jan. 2015 -
Dec. 2018

B.Sc., Computer Science and Engineering, First Class Honors
University of Moratuwa, Sri Lanka
Courses: *Human Computer Interaction, Machine Learning, Advanced Algorithms, Data Mining & Information Retrieval, Distributed Systems*

EXPERIENCE

Jan. 2019 -
Aug. 2024

Old Dominion University, Norfolk, VA
Graduate Research/Teaching Assistant in Computer Science

May 2022 -
May 2023

Learning, Innovation, & Technology Lab, Harvard University, Cambridge, MA
Remote Research Assistant

Jun. - Aug.
2021 & 2022

Los Alamos National Lab, Los Alamos, NM
Summer Research Intern, Institutional Scientific Content Team

Jan. 2018 - Jun.
2018

University of Moratuwa, Sri Lanka
Visiting Instructor in Computer Science

Jun. 2017 -
Dec. 2017

UStocktrade (Pvt) Ltd. Sri Lanka
Software Engineering Intern

AWARDS, HONORS, & SCHOLARSHIPS

1. **Lee Entsminger Outstanding Dissertation Award**, 2024-2025, Old Dominion University
2. **Best Doctoral Symposium Presentation Award**, 2022 Symposium on Eye Tracking Research and Applications (ETRA 2022)

3. **Best Poster Award**, 2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL 2021)
4. **Best Poster Award**, 2020 ACM/IEEE Joint Conference on Digital Libraries (JCDL 2020)
5. **Computer Science, Irwin B. Levinstein Scholarship**, 2020-2021, Old Dominion University (\$2500)
6. **Best Paper Award**, 2019 International Multidisciplinary Engineering Research Conference (MERCon2019)
7. **Computer Science Outstanding Researcher Award**, 2019-2020, Old Dominion University
8. **Dominion Graduate Scholar**, 2019, Old Dominion University (\$20,000)

PUBLICATIONS

UNDER PREPARATION

- 2026 *Jayawardena, Gavindya*, Yasith Jayawardana, Naila Ayala, and Jacek Gwizdka. "Visual Attention and Cognitive Load in Challenging Simulated Flight Landings."
- 2026 *Jayawardena, Gavindya*, Yasith Jayawardana, Naila Ayala, and Jacek Gwizdka. "Dual-Task Demands and Attention in Simulated Flight Landings."

PEER-REVIEWED JOURNAL ARTICLES

- 2025 *Jayawardena, Gavindya*, Yasith Jayawardana, and Jacek Gwizdka. 2025. "Measuring Mental Effort in Real Time Using Pupillometry" *Journal of Eye Movement Research (JEMR)* 18, no. 6: 70.
- 2024 Abeyasinghe, Yasasi, Bhanuka Mahanama, *Gavindya Jayawardena*, Yasith Jayawardana, Mohan Sunkara, Andrew T. Duchowski, Vikas Ashok, and Sampath Jayarathna. "A-disetrac advanced analytic dashboard for distributed eye tracking." *International Journal of Multimedia Data Engineering and Management (IJMDEM)* 15, no. 1 (2024): 1-20.
- 2022 Mahanama, Bhanuka, Yasith Jayawardana, Sundararaman Rengarajan, *Gavindya Jayawardena*, Leanne Chukoskie, Joseph Snider, and Sampath Jayarathna. "Eye movement and pupil measures: A review." *Frontiers in Computer Science* 3 (2022): 733531. **[Editor's Pick]**
- 2020 *Jayawardena, Gavindya*, and Sampath Jayarathna. "Automated filtering of eye movements using dynamic aoi in multiple granularity levels." *International Journal of Multimedia Data Engineering and Management (IJMDEM)* 12, no. 1 (2021): 49-64.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- 2026 *Jayawardena, Gavindya*, Dan Zhang, and Jacek Gwizdka. "Visual Attention and Cognitive Load in Text Relevance Assessment." To appear in Proceedings of the 2026 ACM Symposium on Eye Tracking Research & Applications, The Third - MultipleEYE Workshop (**ETRA '26**)
- 2026 Zhang, Dan, *Gavindya Jayawardena*, and Jacek Gwizdka. "Phase-Dependent Scanpath Dynamics in Map-Based Tasks." To appear in Proceedings of the 2026 ACM Symposium on Eye Tracking Research & Applications (**ETRA '26**)
- 2026 Abeyasinghe, Yasasi, *Gavindya Jayawardena*, Enkelejda Kasneci, and Sampath Jayarathna. "Susceptibility to High-Fidelity Misinformation: An Eye-Tracking Analysis." To appear in Proceedings of the 2026 ACM Symposium on Eye Tracking Research & Applications (**ETRA '26**)
- 2026 *Jayawardena, Gavindya*, Li Shi, and Jacek Gwizdka. "Effects of Working Memory Capacity and Search Task Complexity on Cognitive Load." In Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction and Retrieval (**CHIIR '26**)
- 2026 Dan Zhang, *Gavindya Jayawardena*, and Jacek Gwizdka. "Attention! Rethinking What We Measure in CHIIR Studies." Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction and Retrieval (**CHIIR '26**)
- 2025 *Jayawardena, Gavindya*, Yasith Jayawardana, Yasasi Abeyasinghe, Bhanuka Mahanama, Sampath Jayarathna, and Jacek Gwizdka. "A Real-Time Approach to Capture Ambient and Focal Attention in Visual Search." In Proceedings of the 2025 Symposium on Eye Tracking Research and Applications, pp. 1-7. 2025. (**ETRA '25**)

- 2025 Shi, Li, *Gavindya Jayawardena*, and Jacek Gwizdka. "Pupillometric Analysis of Cognitive Load in Relation to Relevance and Confirmation Bias." In Proceedings of the 2025 ACM SIGIR Conference on Human Information Interaction and Retrieval, pp. 219-230. 2025. (**CHIIR '25**)
- 2024 *Jayawardena, Gavindya*, Vikas Ashok, and Sampath Jayarathna. "Advanced Gaze Analytics Dashboard." In 2024 IEEE International Conference on Information Reuse and Integration for Data Science, pp. 114-119. IEEE, 2024. (**IRI 2024**)
- 2023 Abeysinghe, Yasasi, Bhanuka Mahanama, *Gavindya Jayawardena*, Mohan Sunkara, Vikas Ashok, and Sampath Jayarathna. "Gaze analytics dashboard for distributed eye tracking." In 2023 IEEE 24th International Conference on Information Reuse and Integration for Data Science, pp. 140-145. IEEE, 2023. (**IRI 2023**)
- 2022 *Jayawardena, Gavindya*, Yasith Jayawardana, Sampath Jayarathna, Jonas Högström, Thomas Papa, Deepak Akkil, Andrew T. Duchowski et al. "Toward a real-time index of pupillary activity as an indicator of cognitive load." *Procedia Computer Science* 207 (2022): 1331-1340. (**KES 2022**)
- 2021 Jayawardana, Yasith, *Gavindya Jayawardena*, Andrew T. Duchowski, and Sampath Jayarathna. "Metadata-driven eye tracking for real-time applications." In Proceedings of the 21st ACM Symposium on Document Engineering, pp. 1-4. 2021. (**DocEng 2021**)
- 2021 *Jayawardena, Gavindya*, and Sampath Jayarathna. "Automated filtering of eye gaze metrics from dynamic areas of interest." In 2020 IEEE 21st International Conference on Information Reuse and Integration for Data Science, pp. 67-74. IEEE, 2020. (**IRI 2020**)
- 2020 *Jayawardena, Gavindya*, Anne Michalek, Andrew Duchowski, and Sampath Jayarathna. "Pilot study of audiovisual speech-in-noise (SIN) performance of young adults with adhd." In ACM symposium on eye tracking research and applications, pp. 1-5. 2020. (**ETRA 2020**)
- 2020 Jayawardana, Yasith, Alexander C. Nwala, *Gavindya Jayawardena*, Jian Wu, Sampath Jayarathna, Michael L. Nelson, and C. Lee Giles. "Modeling updates of scholarly webpages using archived data." In 2020 IEEE International Conference on Big Data, pp. 1868-1877. IEEE, 2020. (**Big Data 2020**)
- 2019 De Silva, Senuri, Sanuwani Dayarathna, Gangani Ariyaratne, Dulani Meedeniya, Sampath Jayarathna, Anne MP Michalek, and *Gavindya Jayawardena*. "A rule-based system for ADHD identification using eye movement data." In 2019 Moratuwa engineering research conference, pp. 538-543. IEEE, 2019. (**MERCon 2019**) [**Best Paper Award**]
- 2019 *Jayawardena, Gavindya*, Anne Michalek, and Sampath Jayarathna. "Eye tracking area of interest in the context of working memory capacity tasks." In 2019 IEEE 20th International Conference on Information Reuse and Integration for Data Science, pp. 208-215. IEEE, 2019. (**IRI 2019**)
- 2018 Jayawardana, Yasith, Randil Fernando, *Gavindya Jayawardena*, Dileka Weerasooriya, and Indika Perera. "A full stack microservices framework with business modelling." In 2018 18th International Conference on Advances in ICT for Emerging Regions, pp. 78-85. IEEE, 2018. (**ICTer 2018**)

PEER-REVIEWED BOOK CHAPTERS

- 2019 Michalek, Anne MP, *Gavindya Jayawardena*, and Sampath Jayarathna. "Predicting ADHD using eye gaze metrics indexing working memory capacity." In Computational Models for Biomedical Reasoning and Problem Solving, pp. 66-88. (**IGI Global**), 2019.

DOCTORAL CONSORTIA (not in proceedings)

- 2022 *Jayawardena, Gavindya*. "Introducing a real-time advanced eye movements analysis pipeline." In 2022 Symposium on Eye Tracking Research and Applications, pp. 1-2. 2022. (**ETRA 2022**) [**Best Doctoral Symposium Presentation Award**]
- 2020 *Jayawardena, Gavindya*. "RAEMAP: Real-Time Advanced Eye Movements Analysis Pipeline". In 2020 International Conference on Artificial Intelligence in Medicine, 2020. (**AIME 2020**),
- 2020 *Jayawardena, Gavindya*. "RAEMAP: Real-time advanced eye movements analysis pipeline." In ACM Symposium on Eye Tracking Research and Applications, pp. 1-4. 2020. (**ETRA 2020**)

PEER-REVIEWED POSTERS

- 2022 Mahanama, Bhanuka, *Gavindya Jayawardena*, Yasasi Abeysinghe, Vikas Ashok, and Sampath Jayarathna. "Multidisciplinary reading patterns of digital documents." In 2022 Symposium on Eye Tracking Research and Applications, pp. 1-2. 2022. **(ETRA 2022)**.
- 2021 Mahanama, Bhanuka, *Gavindya Jayawardena*, and Sampath Jayarathna. "Analyzing unconstrained reading patterns of digital documents using eye tracking." In 2021 ACM/IEEE Joint Conference on Digital Libraries, pp. 282-283. IEEE, 2021. **(JCDL 2021) [Best Poster Award]**
- 2020 *Jayawardena, Gavindya*, Sampath Jayarathna, and Jian Wu. "Analyzing the effect of reading patterns using eye tracking measures." In Proceedings of the ACM/IEEE Joint Conference on Digital Libraries in 2020, pp. 517-518. 2020. **(JCDL 2020) [Best Poster Award]**

PEER-REVIEWED WORKSHOPS

- 2024 Spina, Damiano, Jacek Gwizdka, Kaixin Ji, Yashar Moshfeghi, Javed Mostafa, Tuukka Ruotsalo, Min Zhang, Adnan Ahmad, Sara Fahad Dawood Al Lawati, Nattapat Boonprakong, Nishani Fernando, Jiaman He, Orland Hoeber, *Gavindya Jayawardena*, Boon-Giin Lee, Haiming Liu, Matthew Pike, Abbas Pirmoradi, Bahareh Nakisa, Mohammad Naim Rastgoo, Flora D. Salim, Fletcher Scott, Shuoqi Sun, Huimin Tang, Dave Towey, Max L. Wilson. "Report on the 3rd Workshop on NeuroPhysiological Approaches for Interactive Information Retrieval at SIGIR CHIIR 2025." In *ACM SIGIR Forum*, vol. 59, no. 1, pp. 1-43. New York, NY, USA: ACM, 2025. **(NeuroPhysIIR 2025)**

OTHER ACADEMIC CONTRIBUTIONS (Not peer-reviewed, not in proceedings)

- 2026 *Gavindya Jayawardena*, Li Shi, Dan Zhang, Kai-Yu (Kelly) Chang, Cecelia Albright, Hsin-Yu (Joyce) Chang, Pranati Kompella, Jacek Gwizdka. "Do AI Copilots Support Learning? An Eye-Tracking Study of Attention and Difficulty." Poster presented at 2026 iSchool Research Showcase, Austin, Texas (2026)
- 2026 *Gavindya Jayawardena*, Li Shi, Jacek Gwizdka. "How Working Memory Shapes Cognitive Load and Emotion in Interactive Search." Poster presented at 2026 iSchool Research Showcase, Austin, Texas (2026)
- 2026 Dan Zhang, *Gavindya Jayawardena*, Jacek Gwizdka. "Revisiting "Attention" in CHIIR Research: A Review of Definitions and Measurements (2016–2025)." Poster presented at 2026 iSchool Research Showcase, Austin, Texas (2026)
- 2024 Shi, Li, *Gavindya Jayawardena*, Jacek Gwizdka. "Relevance, Confirmation Bias, and Cognitive Load: An Eye-tracking Study." Poster presented at 2024 iSchool AI Showcase, Austin, Texas (2024)
- 2023 *Jayawardena, Gavindya*, Sampath Jayarathna, and Yi He. "ADHD Prediction Through Analysis of Eye Movements With Graph Convolution Network." In 2023 Graduate Research Achievement Day, College of Sciences, ODU (2023).
- 2023 Owens, James, *Gavindya Jayawardena*, Yasasi Abeysinghe, Vikas G. Ashok, and Sampath Jayarathna. "Main Sequence Relationships, Task Load, and Cognitive Load." In Undergraduate Research Symposium, College of Sciences, ODU (2023).
- 2023 Abeysinghe, Yasasi, *Gavindya Jayawardana*, Autumn Woodson, Efe Bozkir, Enkelejda Kasneci, Andrew Duchowski, and Sampath Jayarathna. "Exploring Human Perception while Reading Fake and Real News Articles." In 2023 Graduate Research Achievement Day, College of Sciences, ODU (2023).
- 2021 *Gavindya Jayawardena*, Sampath Jayarathna, and Jian Wu. "Analysis of Reading Patterns of Scientific Literature using Eye-Tracking Measures." In 2021 Graduate Research Achievement Day, College of Sciences, ODU (2021).
- 2020 *Gavindya Jayawardena*, Anne Michalek, and Sampath Jayarathna, "Pilot Study of Audiovisual Speech-In-Noise (SIN) Performance of Adolescence with ADHD Using Eye-Tracking", ACM CAPWIC, Newport News, VA (2020).

INVITED TALKS

- 1. "Effects of Working Memory Capacity and Search Task Complexity on Cognitive Load." Postdoc Colloquia, School of Information, The University of Texas at Austin, 2025

2. "Visual Processing in Action: The Dynamics of Attention Allocation." Postdoctoral Research Expo, School of Information, The University of Texas at Austin, 2024
3. "Introduction to Numpy and Pandas". Coastal Adaptation and Resilience Education (CARE) Summer Camp, Computer Science Department, Old Dominion University, 2023
4. "RAEMAP: Real-time Advanced Eye Movements Analysis Pipeline". Student Presenter at Web Science & Digital Libraries Research Group Expo (WS-DL Expo), Computer Science Department, Old Dominion University, 2023
5. "Eye Tracking for Predicting ADHD". CS Summer Research Workshop, Old Dominion University, 2019
6. "Eye-Tracking for Assessing Speech-In-Noise Performance of Adults with ADHD". Guest Lecture for CS395 - Research Methods in Data & Web Science, Old Dominion University, 2020
7. "Introduction to Data Wrangling". Guest Lecture for STRS: Student ThinSat Research Summer Camp, Old Dominion University, 2020

TEACHING EXPERIENCE

Fall 2025	Guest Lecturer - INF 385T: Biometrics for UX, The University of Texas at Austin
Summer 2024	Course Instructor - CS115: Introduction to Python, Old Dominion University
Spring 2021	Teaching Assistant - CS250: Problem Solving and Programming II, Old Dominion University
Fall 2020	Teaching Assistant - CS334: Computer Architecture, Old Dominion University
Fall 2020	Teaching Assistant - CS450: Database, Old Dominion University
Spring 2020	Teaching Assistant - CS441/541: App Development for Smart Devices, Old Dominion University
Spring 2018	Teaching Assistant - CS2963: Presentation Skills, University of Moratuwa
Summer 2023	Workshop Instructor - CARE Summer High School Workshop, Old Dominion University
Summer 2020	Workshop Instructor - STRS: Student ThinSat Research Summer Camp, Old Dominion University

SERVICE

Graduate Mentoring	IX Lab, School of Information, The University of Texas at Austin, 2024-present
Undergraduate Mentoring	NIRDS Lab, Old Dominion University, 2021-2024 NSF Research Experiences for Undergraduates (REU 2023), Computer Science Department, Old Dominion University, 2023
Reviewing and Editing	Frontiers in Psychology (Cognition), 2026 Acta Psychologica, 2026 Frontiers in Virtual Reality, 2025 (IWC) Interacting with Computers - Oxford Academic, 2025 (ETRA) Eye Tracking Research & Applications Symposium, 2025, 2024, 2020 (OzCHI) Australian Conference on Human-Computer Interaction, 2024 (CHI) Conference on Human Factors in Computing Systems, 2024 (IUI) Intelligent User Interfaces, 2020, 2021, 2022 (JCDL) ACM/IEEE-CS Joint Conference on Digital Libraries, 2025, 2024, 2022, 2020 (AH) Augmented Human, 2020 (CIKM) Conference on Information and Knowledge Management, 2022 (CHIIR) ACM Conference on Human Information Interaction & Retrieval, 2021 IEEE BigData, 2021, 2023
Leadership	President of ACM Student Chapter-Women, Old Dominion University, 2022-2023 President of ACM Student Chapter, University of Moratuwa, 2018
Volunteering	IEEE IRI 2024 & IEEE MIPR 2024 Conference, Student Volunteer 2nd Annual JRMF Math & Computer Science Festival, Old Dominion University 34th Great Computer Challenge (Grades 6-12), Old Dominion University STEAM on Spectrum event, VMASC, VA, 2019

RESEARCH-BASED & TECHNICAL BLOG ARTICLES

1. Effect of Reading Patterns of Novice Researchers using Eye Tracking
<https://ws-dl.blogspot.com/2020/04/2020-04-25-effect-of-reading-patterns.html>
This article explains how novice researchers' eye movements change during a research papers reading task.
2. Lab Streaming Layer (LSL) Tutorial for Windows
<https://ws-dl.blogspot.com/2019/07/2019-07-15-lab-streaming-layer-lsl.html>
This article explains how to install open source LSL and stream eye movement data from PupilLabs Core eye tracker to NeuroPype academic edition.
3. Use of Cognitive Memory to Improve the Accessibility of Digital Collections
<https://ws-dl.blogspot.com/2019/06/2019-06-19-use-of-cognitive-memory-to.html>
This article explains a study conducted to investigate eye gaze metrics collected during Reading SPAN task to differentiate the performance of adults with and without ADHD, and how to generalize that study to improve content accessibility for the people with learning disabilities without overloading their cognitive memory.
4. Real-time Header Extraction from Scientific PDF Documents: Summer Research Internship Experience at Los Alamos National Laboratory
<https://ws-dl.blogspot.com/2021/09/2021-09-23-real-time-header-extraction.html>
This article explains my 2021 Summer research internship experience at Los Alamos National Laboratory.
5. PDFServer - Our Summer Internship at LANL
<https://ws-dl.blogspot.com/2022/12/2022-12-21-pdfserver-our-summer.html>
This article explains my 2022 Summer research internship experience at Los Alamos National Laboratory.
6. ADHD Prediction Through Analysis of Eye Movements With Graph Convolution Network
<https://ws-dl.blogspot.com/2022/06/2022-06-18-adhd-prediction-through.html>
This article discusses how I utilized graphs generated using eye-tracking data for the diagnosis of ADHD.

OTHER AWARDS & SCHOLARSHIPS

1. **Outstanding Service Award**, IEEE IRI 2024 & MIPR 2024, Student Volunteer
2. **Dean's List** - Semester 1, 6, 7, 8, University of Moratuwa, Sri Lanka
3. **Mahapola Higher Education Scholarship** (4 academic years) - 20014/2018
University of Moratuwa, Sri Lanka

MEMBERSHIPS

Member, ACM

Member, SIGCHI

Past Member, The Honor Society of Phi Kappa Phi

Past Student member, Association for Computing Machinery – Women (ACM-W)

Past Student member, IEEE Computer Society